

ST 3007 – Operational Research

Tutorial 3

Transportation Problems

Q1) Four factories A, B, C and D are required to manufacture and supply a specific product for four stores P, Q, R and S. Production capacities in each factory and demand at each store are given in table 1 below, while Unit transportation cost in Rupees from each factory to each store is given in table 2.

Table 1

Factory	A	B	C	D
Production capacity (units)	50	70	30	50
Store	P	Q	R	S
Demand	25	35	105	35

Table 2

Store		P	Q	R	S
Factory	A	2	4	8	11
	B	10	8	7	5
	C	13	3	9	12
	D	4	6	8	3

- Construct this as a standard Transportation problem in tabular form.
- Using the least cost method or any other method, find the initial solution.
- Check for optimality of the solution obtained in part (b) using any appropriate method.
- Has the optimal solution been obtained? If the answer is no, improve it until the optimal solution is obtained. If the answer is yes, justify your answer.
- Evaluate the optimum transportation cost.

Q2) A computer manufacturer has two factories, one each at cities F1 and F2 and three warehouses located at cities W1, W2 and W3. The following table gives the monthly availability of the factories, monthly requirements at the warehouses and the per unit transportation costs

	W1	W2	W3	Availability
F1	4	7	3	5000
F2	9	4	8	10000
Requirement	4000	8000	3000	

- Determine the initial shipping plan by using the North-West corner method.
- Check for optimality of the solution obtained in part(a)
- Evaluate the optimum transportation schedule and the optimum cost.
- Are there multiple optimal solutions? Give reasons for your answer. If the answer is yes, write down another alternate optimal solution.

Q3) A Company has 3 plants, namely A, B and C, which produce 300 kg, 200 kg, and 500 kg respectively of a particular product per day. The productions cost per kilogram in each of these plants is

the same and is equal to Rs.60. Four consumers have placed orders for the product on the basis given in the following table.

Consumer	Requirement	Price/kg
1	400	73
2	220	73
3	220	73
4	160	73

Transportation costs in Rupees/kg from the plants to consumers are given in the table below.

	1	2	3	4
A	3	8	3	6
B	8	11	9	12
C	5	6	2	8

- Represent this as a transportation tableau
- Apply the North-West corner method to obtain the initial feasible transportation schedule.
- Evaluate each empty cell to test for optimality
- If optimality has been reached, evaluate the minimum transportation cost. If not, improve the solution until optimality is reached and find the minimum transportation cost.
- Interpret the solution you obtained from part (c), so that any ordinary person would understand the transportation schedule. State any special features of the solution if there are any.

Non-Traditional Transportation models (production-inventory Control)

Q4) Boralis manufactures backpacks for serious hikers. The demand for its product during the peak period of March to June of each year is 100, 200, 180, and 300 units, respectively. The company uses part-time labor to accommodate fluctuations in demand. It is estimated that Boralis can produce 50, 180, 280, and 270 units in March through June. The current month's demand may be satisfied in one of three ways.

- The current month's production at the cost of \$40 per pack.
- Surplus production in an earlier month at an additional holding cost of \$.50 per pack per month
- Surplus production in a later month (back-ordering) at an additional penalty cost of \$2.00 per pack per month.

Determine the optimal production schedule for the four months.